Triple Integration

$$\int_{x=a}^{b}\int_{y=f_1(x)}^{f_2(x)}\int_{z=g_1(x,y)}^{g_2(x,y)} f(x,y,z)\,dz\,dy\,dx$$

$\cancel{x^2+y^2}$

$\cancel{x^2-y^2}$

$\cancel{\frac{1}{x^2}+\frac{1}{y^2}}$

$\cancel{x^2\left(-\frac{y^3}{3}\right)}$

1. $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dxdydz}{\sqrt{(1-x^2-y^2-z^2)}}$

$$= \int_{x=0}^{1} \int_{y=0}^{\sqrt{1-x^2}} \int_{z=0}^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{(1-x^2-y^2-z^2)}}\, dz\, dy\, dx$$

$$1-x^2-y^2 = a^2$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{a} \frac{dz}{\sqrt{(a^2-z^2)}}\, dy\, dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1}\left(\frac{z}{a}\right) \right]_0^a dy\, dx \quad \left(\sin^{-1}(1) - \sin^{-1} 0 \right)$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\pi/2\right) dy\, dx$$

$$= \pi/2 \int_0^1 (y)_0^{\sqrt{1-x^2}} dx = \pi/2 \int_0^1 \sqrt{1-x^2}\, dx$$

$$= \pi/2 \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1}(x) \right]_0^1$$

$$= \pi/2 \left[\frac{1}{2}(0) + \frac{1}{2}\, \pi/2 \right] = \frac{\pi^2}{8} //$$

$$\int \sqrt{a^2-x^2}\, dx$$

$$= \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$$

2. $\int_0^2 \int_0^x \int_0^{2x+2y} e^{x+y+z}\,dz\,dy\,dx$

$\int e^{ax}\,dx = \frac{e^{ax}}{a}$

$\hookrightarrow \underline{e^{x} \cdot e^{y}} \left(e^{z} \right)$

$\left(e^{z} \right)_{0}^{2x+2y} = \left(e^{2x+2y} - 1 \right)$

Ans $= \frac{e^{12}}{8} - \frac{e^{4}}{2} - \frac{e^{6}}{9} + e^{2} - \frac{4}{9}$

$$\int_0^{\pi} 2d\theta \int_0^{a(1+cos\theta)} r\,dr \int_0^{h} \left|1-\frac{r}{a(1+cos\theta)}\right| dz$$

$$= 2\int_0^{\pi}\int_0^{a(1+\cos\theta)}\int_0^{h} r\left(1-\frac{r}{a(1+\cos\theta)}\right) dz\,dr\,d\theta$$

$$= 2\int_0^{\pi}\int_0^{a(1+\cos\theta)}\left(r-\frac{r^2}{a(1+\cos\theta)}\right)\underbrace{(z)_0^h}_{h}\,dr\,d\theta$$

$$= 2h\int_0^{\pi}\int_0^{a(1+\cos\theta)}\left(r-\frac{r^2}{a(1+\cos\theta)}\right)dr\,d\theta$$

$$= 2h\int_0^{\pi}\left(\frac{r^2}{2}-\frac{1}{a(1+\cos\theta)}\frac{r^3}{3}\right)_0^{a(1+\cos\theta)}d\theta$$

$$= 2h\int_0^{\pi}\left[\frac{a^2(1+\cos\theta)^2}{2}-\frac{1}{\cancel{a(1+\cos\theta)}}\frac{a^3(1+\cos\theta)^3}{3}\right]d\theta$$

$$= 2h\int_0^{\pi} a^2(1+\cos\theta)^2\left(\frac{1}{2}-\frac{1}{3}\right)d\theta$$

$$= \frac{2ha^2}{6}\int_0^{\pi}\left(1+2\cos\theta+\cos^2\theta\right)d\theta$$

$$= \frac{ha^2}{3}\int_0^{\pi}\left(1+2\cos\theta+\frac{1+\cos2\theta}{2}\right)d\theta$$

$$= \frac{ha^2}{3}\int_0^{\pi}\left(\frac{3}{2}+2\cos\theta+\frac{\cos2\theta}{2}\right)d\theta$$

$$= \frac{ha^2}{3}\left[\frac{3}{2}\theta+2\sin\theta+\frac{\sin2\theta}{4}\right]_0^{\pi}$$

$$= \frac{ha^2}{3}\left[\frac{3}{2}\pi+0\right] = \frac{ha^2\pi}{2}$$

7. $\int_{-2}^{2}\int_{-\sqrt{4-x^2}/2}^{\sqrt{4-x^2}/2}\int_{x^2+3y^2}^{8-x^2-y^2} dz\,dy\,dx$

8. $\int_0^a \int_0^a \int_0^a (yz + zx + xy)\,dx\,dy\,dz.$

$$\int_0^a \int_0^a \left(xyz + \frac{zx^2}{2} + \frac{yx^2}{2} \right)_0^a dy\,dz$$

$$\frac{3}{4} a^5$$

TYPE II : WHEN THE REGION OF INTEGRATION IS BOUNDED BY PLANES

1. Evaluate $\iiint x^2yz\,dx\,dy\,dz$ throughout the volume bounded by the planes $x = 0, y = 0, z = 0, \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

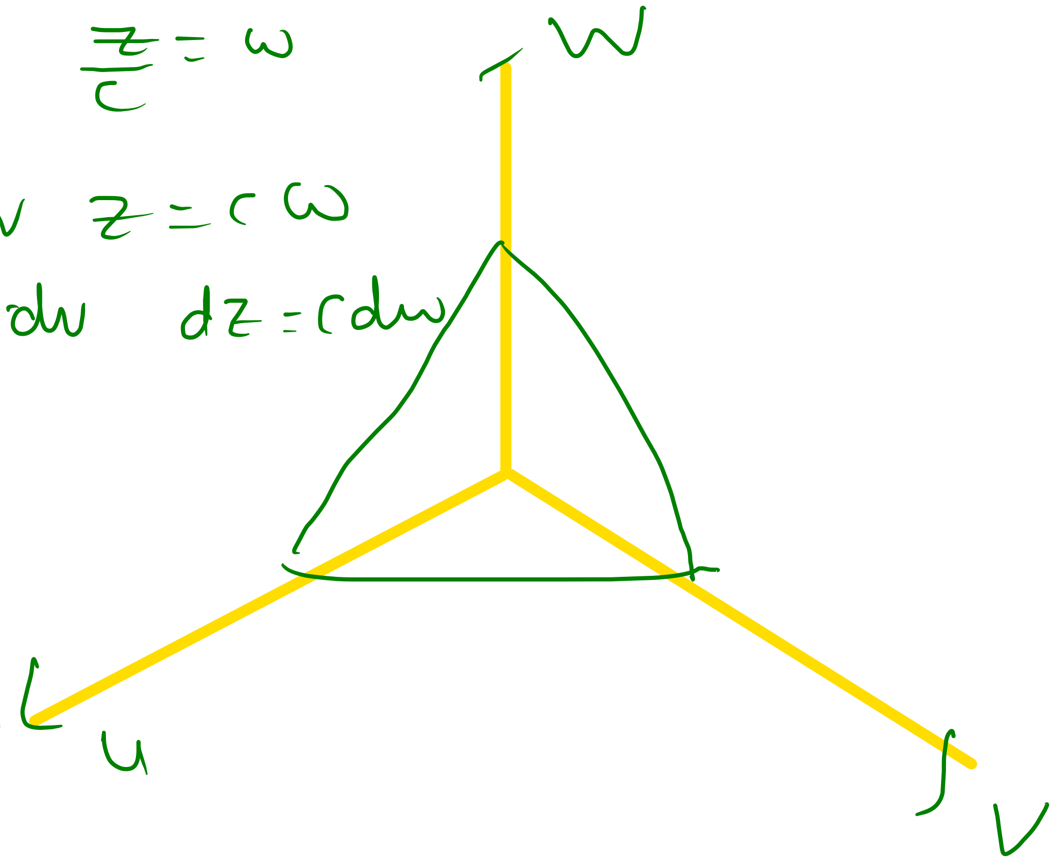
$x = 0$ | $u = 0$
$y = 0$ | $v = 0$
$z = 0$ | $w = 0$

$u + v + w = 1$

$\frac{x}{a} = u \quad \frac{y}{b} = v \quad \frac{z}{c} = w$

$x = au \quad y = bv \quad z = cw$

$dx = a\,du \quad dy = b\,dv \quad dz = c\,dw$



$$\int_0^1 \int_0^{1-u} \int_0^{1-u-v} a^2u^2\, bv\, cw\; a\,du\; b\,dv\; c\,dw$$

$$\int_0^1 \int_0^{1-u} \int_0^{1-u-v} a^3 b^2 c^2\; u^2 v w \; dw\; dv\; du$$

3. Evaluate $\iiint \frac{dx\,dy\,dz}{(1+x+y+z)^3}$ over the volume of the tetrahedron $x = 0, y = 0, z = 0, x + y + z = 1$

$x+y+z=1$

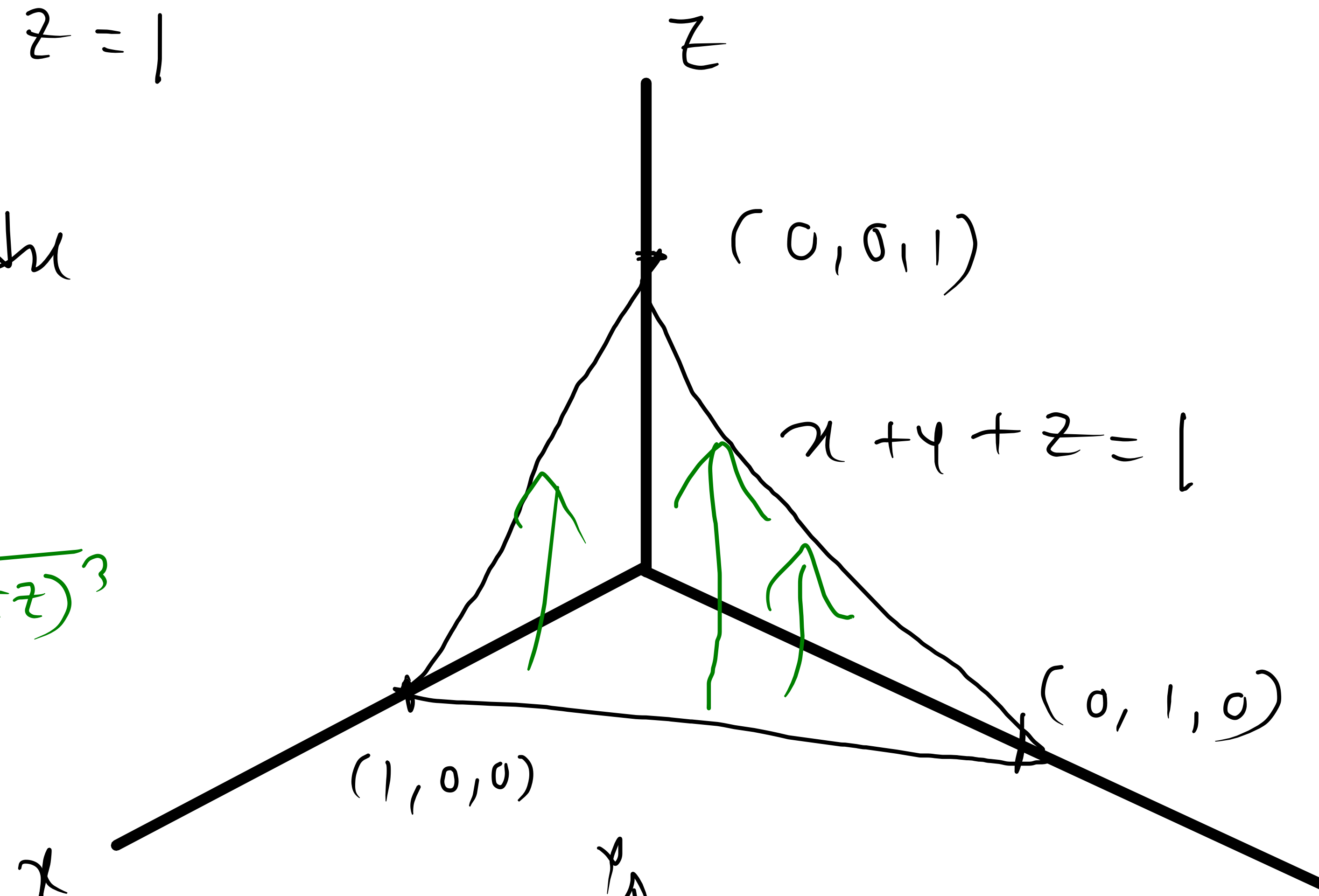
$$\int_{x=0}^{1}\int_{y=0}^{1-x}\int_{z=0}^{1-x-y}\frac{1}{(\underbrace{1+x+y}_{a}+z)^3}\,dz\,dy\,dx$$

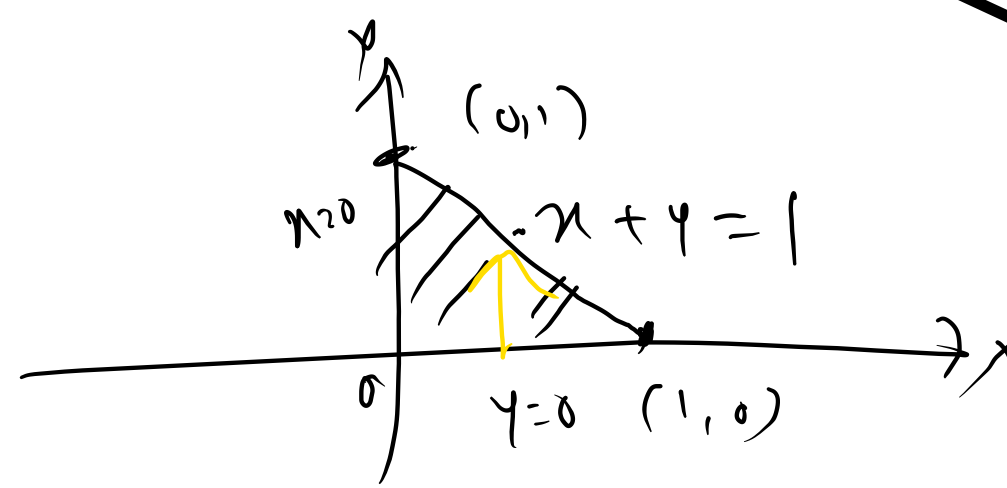
$$=\frac{1}{2}\left[\log 2-\frac{5}{8}\right]$$

$$\frac{1}{(a+z)^3}$$

$$\int (a+z)^{-3}$$

$$\frac{(a+z)^{-2}}{-2}$$





6. Evaluate the integral $\iiint_{v} xyz^2 \, dv$ over the region bounded by the planes $x = 0, x = 1, y = -1, y = 2,$ $z = 0, z = 3$

Cylindrical co-ordinates.

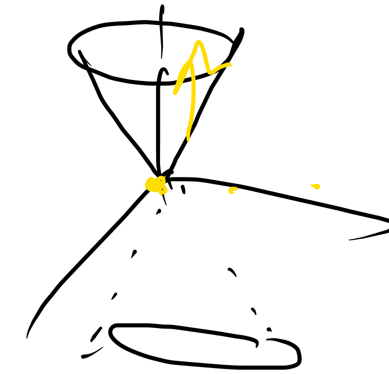
$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$
$$dz\,dy\,dx = r\,dz\,dr\,d\theta$$

Cylinder

$$x^2+y^2=a^2$$

Cone

$$x^2+y^2=z^2$$



$z>0$

paraboloid

$$x^2+y^2=z$$

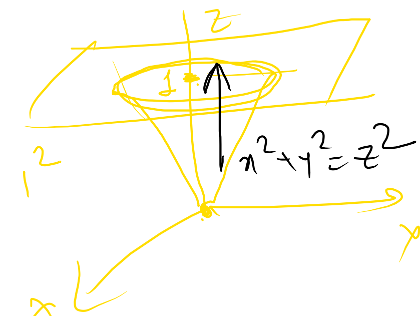
TYPE IV : WHEN THE REGION OF INTEGRATION IS BOUNDED BY A CONE OR A CYLINDER OR A PARABOLOID.

1. $\iiint \sqrt{x^2+y^2}\; dx\, dy\, dz$ over the volume bounded by the right circular cone $x^2 + y^2 = z^2, z > 0$ and the planes $z = 0$ and $z = 1$.

for $z = 1$

$x^2+y^2=z^2$

becomes $x^2+y^2=1^2$



$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$dz\, dy\, dx = r\, dz\, dr\, d\theta$

$$\int_0^{2\pi}\int_0^1\int_r^1 r\, r\, dz\, dr\, d\theta$$

$= \pi/6$

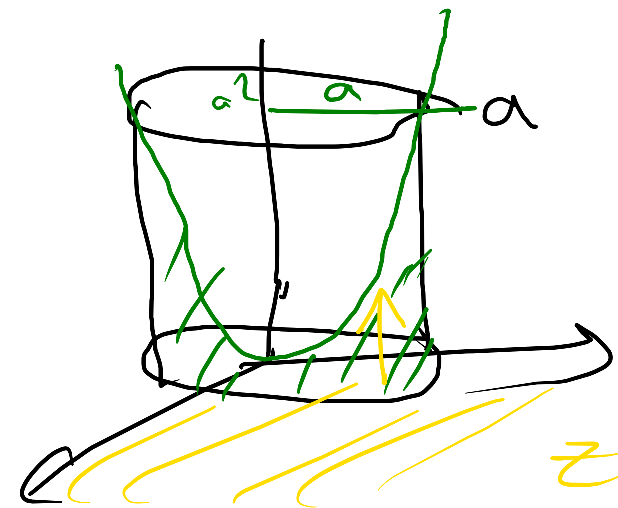
$x^2+y^2=z^2$

$r^2=z^2$

$z=r$

2. $\iiint z^2\, dx\, dy\, dz$ over the volume bounded by the cylinder $x^2+y^2=\cancel{a}^2$ and the paraboloid $x^2+y^2=z$ and the plane $z=0$

$z = a^2$



$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$x^2+y^2=z$

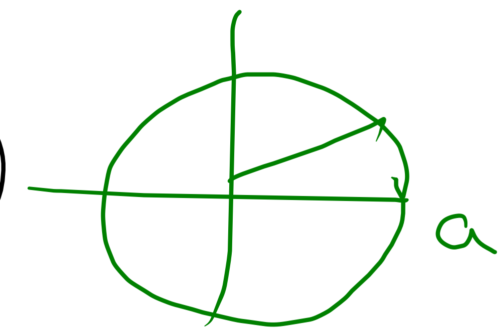
$r^2 = z$

$z \geq 0$

$dz\,dy\,dx = r\,dz\,dr\,d\theta$

$$\int_0^{2\pi}\int_0^{a}\int_0^{r^2} z^2\, r\, dz\, dr\, d\theta$$

$= \dfrac{\pi a^8}{12}$



$x^2+y^2=2$

$x^2+y^2=z$

$\cancel{z=0}$ & $z=2$

4. Evaluate $\iiint_V (x^2 + y^2)\, dV$ where V is the solid bounded by the surface $x^2 + y^2 = z^2$ and the planes $z = 0, z = 2$

Spherical Co-ordinates

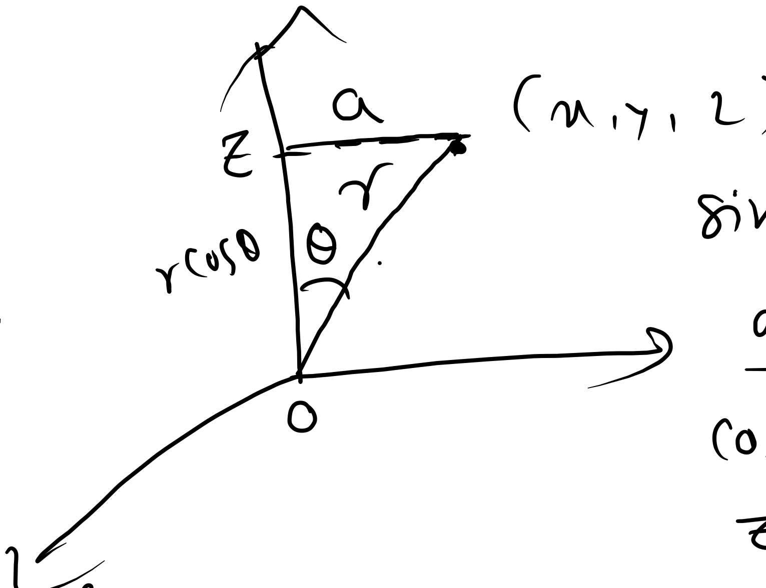
$$x^2+y^2+z^2=a^2 \longrightarrow \text{Sphere}$$

(r, ϕ, θ)

r → radiun in 3dim.

θ → angle from +ve z axis.

ϕ → is angle on xy plane
how the diagram is rotated around
z axis



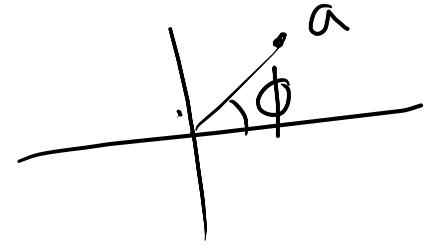
$\sin\theta = \frac{a}{r}$

$\underline{a = r\sin\theta}$

$\cos\theta = \frac{z}{r}$

$\underline{\underline{z = r\cos\theta.}}$

$$\left\{\begin{aligned} x &= \underline{\underline{r\sin\theta}}\cos\phi \\ y &= \underline{\underline{r\sin\theta}}\sin\phi \\ z &= r\cos\theta \end{aligned}\right.$$



$x = r\cos\theta$
$y = r\sin\theta$
$z = z$

$x = a\cos\phi$
$y = a\sin\phi$

$$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi.$$

$$\left.\begin{aligned} 0 &< r < \infty \\ 0 &< \theta < \pi \\ 0 &< \phi < 2\pi \end{aligned}\right\}$$

$$x^2 + y^2 + z^2 = r^2$$

Spherical co-ordinates (θ, ϕ, r)

θ → angle from +ve z axis

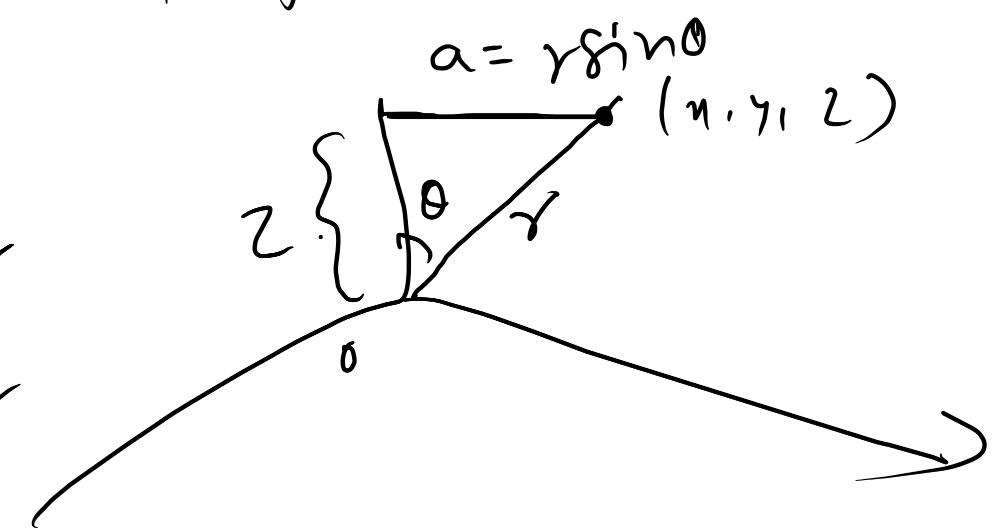
ϕ → angle from +ve x-axis after proj. diag. on xy plane

r → radius in 3 Dim.

$$x = r \sin\theta \cos\phi$$
$$y = r \sin\theta \sin\phi$$
$$z = \underline{r \cos\theta}$$

$$dz\, dy\, dx = r^2 \sin\theta \; dr\, d\theta\, d\phi$$

$$\sin\theta = a/r$$
$$\cos\theta = z/r$$



$$\left\{ \begin{array}{l} 0 \le \phi \le 2\pi \\ 0 \le \theta \le \pi \end{array} \right\}$$

$$A = \int\int 1 \, dy \, dx$$

$$\int\int\int (z) \, dz \, dy \, dx$$

$$V = \int_0 \int_0 \int_0^{f(x,y)} (z) \quad dy \, dx$$

$$\int_0^{1} \int_0^{f(y)} f(x,y)$$

$$V = \int\int\int 1 \, dz \, dy \, dx$$

TYPE III : WHEN THE REGION OF INTEGRATION IS NOT BOUNDED BY PLANES, BUT BY SPHERE, ELLIPSOID ETC.

Evaluate the following integrals.

1. $\iiint_v \frac{dx\,dy\,dz}{(1+x^2+y^2+z^2)^2}$ where V is the volume in the first octant.

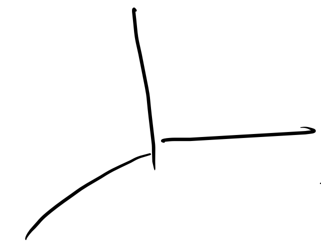
$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$

$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$

$0 \le \theta \le \pi/2$

$0 \le \phi \le \pi/2$

$0 \le r \le \infty$



$\frac{\sin^2 t}{\cos^2 t}\cos^2 t$

$\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} = \beta(m,n)$

$I = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^{\infty} \frac{r^2\sin\theta\,dr\,d\theta\,d\phi}{(1+r^2)^2}$

$r = \tan t$

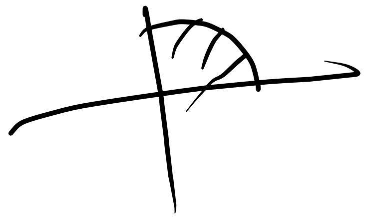
$dr = \sec^2 t\,dt$

$r : 0 \to \infty$

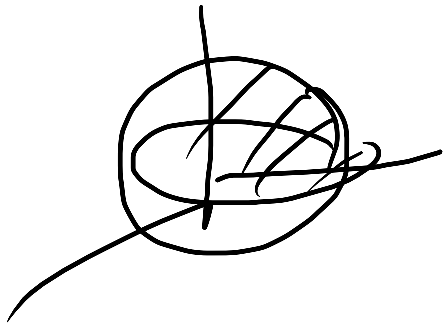
$t : 0 \to \pi/2$

$\int_0^{\pi/2}\int_0^{\pi/2}\sin\theta\int_0^{\pi/2} \frac{\tan^2 t}{\sec^4 t}\sec^2 t\,dt\,d\theta\,d\phi$

$= \frac{\pi^2}{8}$



2. $\iiint (x^2+y^2+z^2)\,dx\,dy\,dz$ over the first octant of the sphere $x^2 + y^2 + z^2 = a^2$



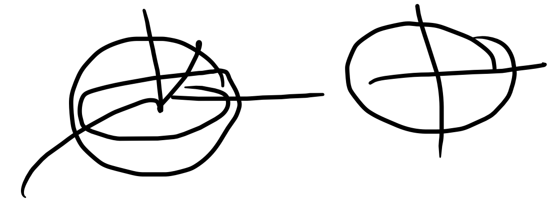
$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$

$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$

$$\int_0^{\pi/2}\int_0^{\pi/2}\int_0^{a} r^2\; r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= \frac{\pi a^5}{10}$$

$x^2+y^2+z^2=a^2$



6. $\iiint \frac{z^2\,dx\,dy\,dz}{x^2+y^2+z^2}$ over the volume of the sphere $x^2+y^2+z^2=2$

$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$

$$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{\sqrt{2}} \frac{r^2\cos^2\theta}{\cancel{r^2}}\,\cancel{r^2}\sin\theta\,dr\,d\theta\,d\phi$$

$$\frac{8\sqrt{2}\,\pi}{9}$$

10. $\iiint_V \frac{z^2}{x^2+y^2+z^2}\,dx\,dy\,dz$ where V is the volume bounded by the sphere $x^2+y^2+z^2=z$

$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$

$$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$x^2+y^2+z^2-z=0$$

$$x^2+y^2+z^2-z+\frac{1}{4}=\frac{1}{4}$$

$$x^2+y^2+\left(z-\frac{1}{2}\right)^2=\frac{1}{2^2}$$

center $(0,0,\tfrac{1}{2})$

rad $\tfrac{1}{2}$

$$\int_0^{2\pi}\int_0^{\pi/2}\int_0^{\cos\theta} \frac{r^2\cos^2\theta}{\cancel{r^2}}\,\cancel{r^2}\sin\theta\,dr\,d\theta\,d\phi = \frac{\pi}{9}$$

$\cancel{2\pi\,\beta\left(\tfrac{3}{2},1\right)\,\tfrac{2\sqrt{2}}{3}}$

$\cancel{\frac{8\sqrt{2}\pi}{9}}$

$$x^2+y^2+z^2=z$$

$$\Rightarrow r^2 = r\cos\theta$$

11. $\iiint_V \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{3/2}}$ where V is the volume bounded by the spheres $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2, (b > a)$

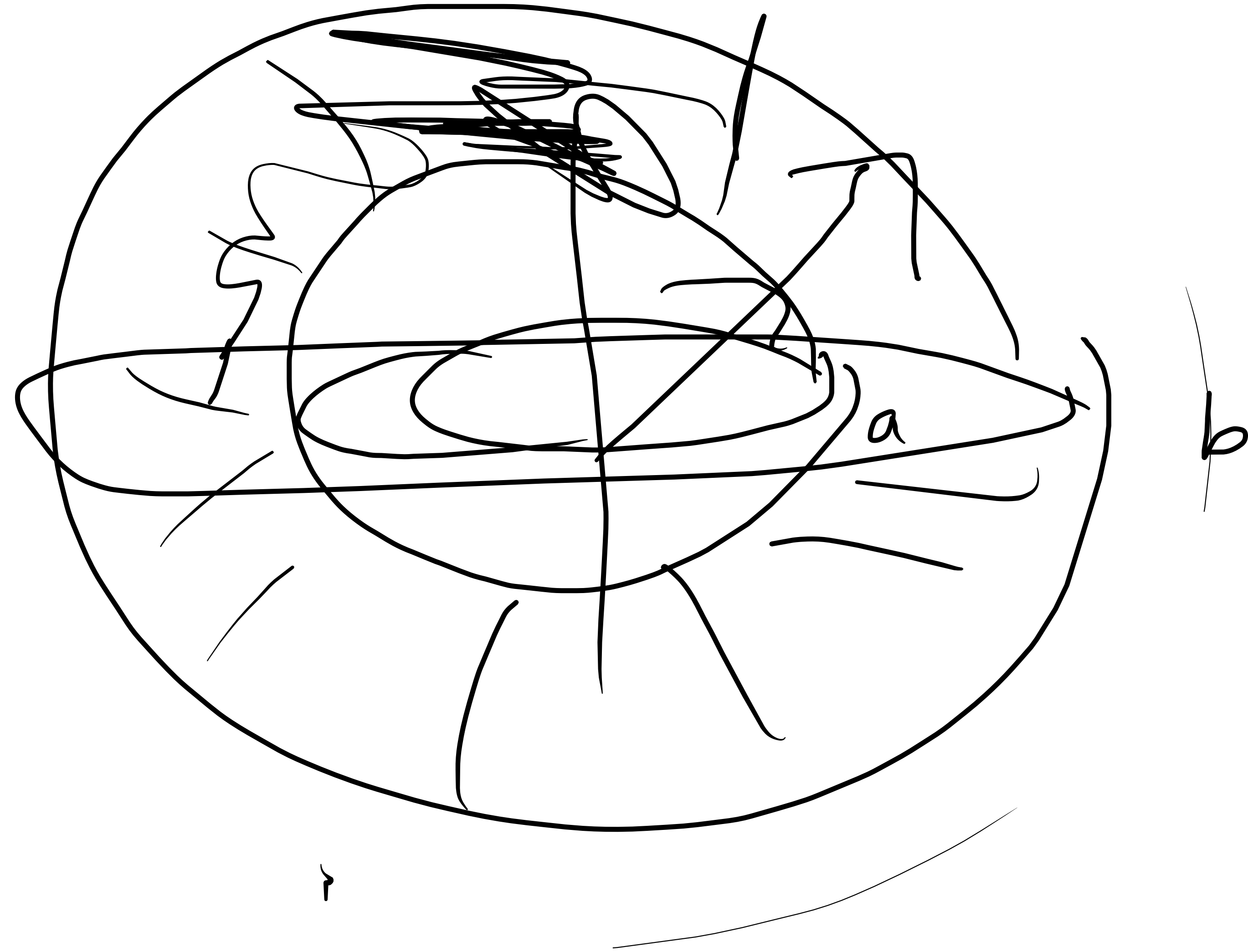
$$\left\{\begin{array}{l} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{array}\right.$$

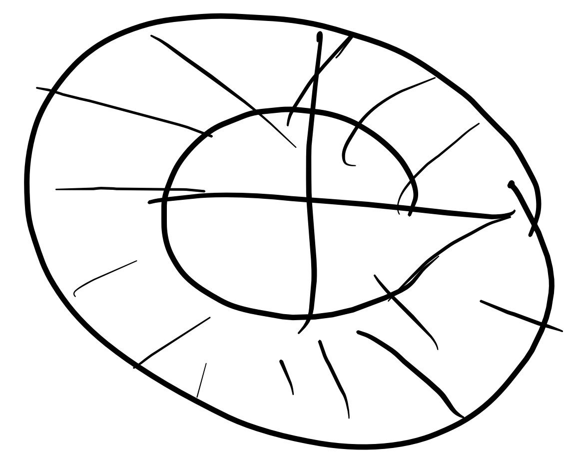
$$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$$

ϕ :

θ :

γ :





Volume

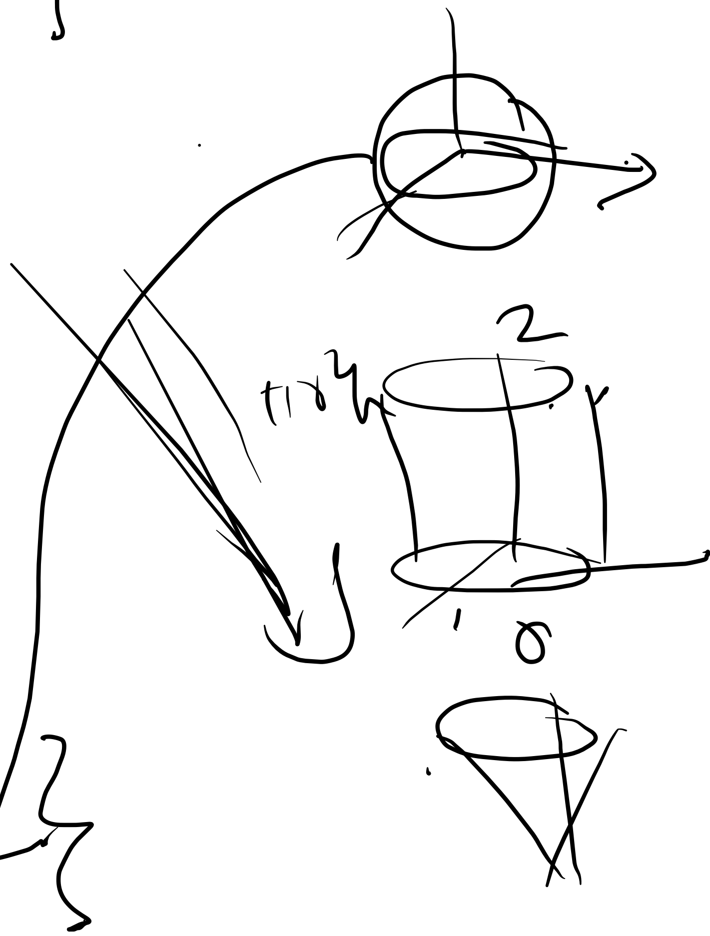
Volume

$V = \iiint 1 \, dz \, dy \, dx$

cylindrical.

$V = \iiint r \, dz \, dr \, d\theta$

$x = r\cos\theta$
$y = r\sin\theta$
$z = z$



spherical.

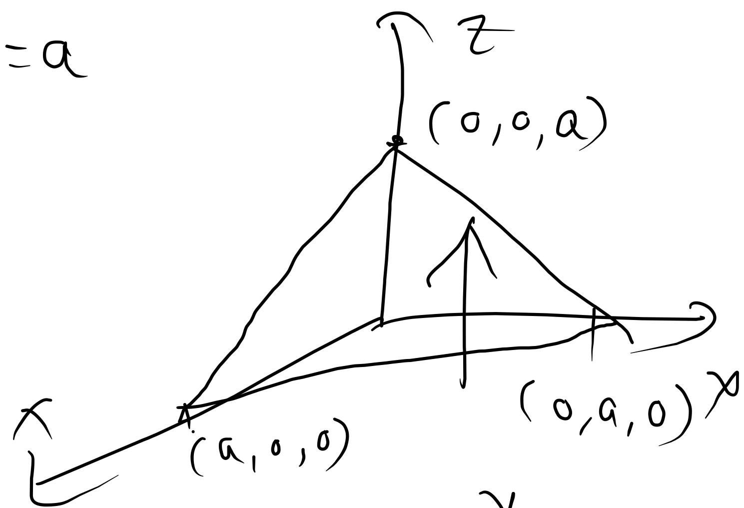
$$\begin{cases} x = r\sin\theta \cos\phi \\ y = r\sin\theta \sin\phi \\ z = r\cos\theta \end{cases}$$

$dz \, dy \, dx = r^2 \sin\theta \, dr \, d\theta \, d\phi$

$V = \iiint r^2 \sin\theta \, dr \, d\theta \, d\phi$

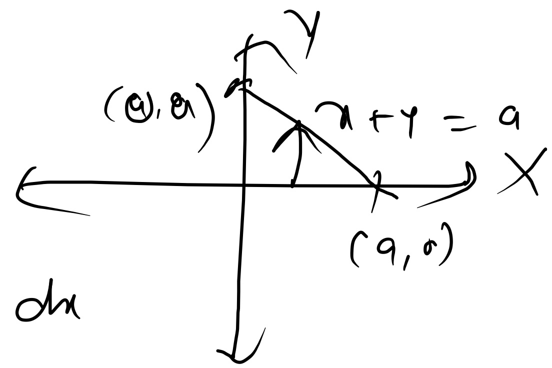
1. Find the volume of the tetrahedron bounded by the planes $x = 0, y = 0, z = 0$ and $x + y + z = a$.

$x + y + z = a$



$$V = \int_0^a \int_0^{a-x} \int_0^{a-x-y} 1 \, dz \, \underline{\underline{dy \, dx}}$$

$$= \int_0^a \int_0^{a-x} (a - x - y) \, dy \, dx$$



$$= \int_0^a \left[ay - xy - \frac{y^2}{2} \right]_0^{a-x} dx$$

$$= \int_0^a \left[a(a-x) - x(a-x) - \frac{(a-x)^2}{2} \right] dx$$

$$= \int_0^a \left[\underline{a^2} - \underline{\underline{ax - ax}} + x^2 - \frac{(a^2 - 2ax + x^2)}{2} \right] dx$$

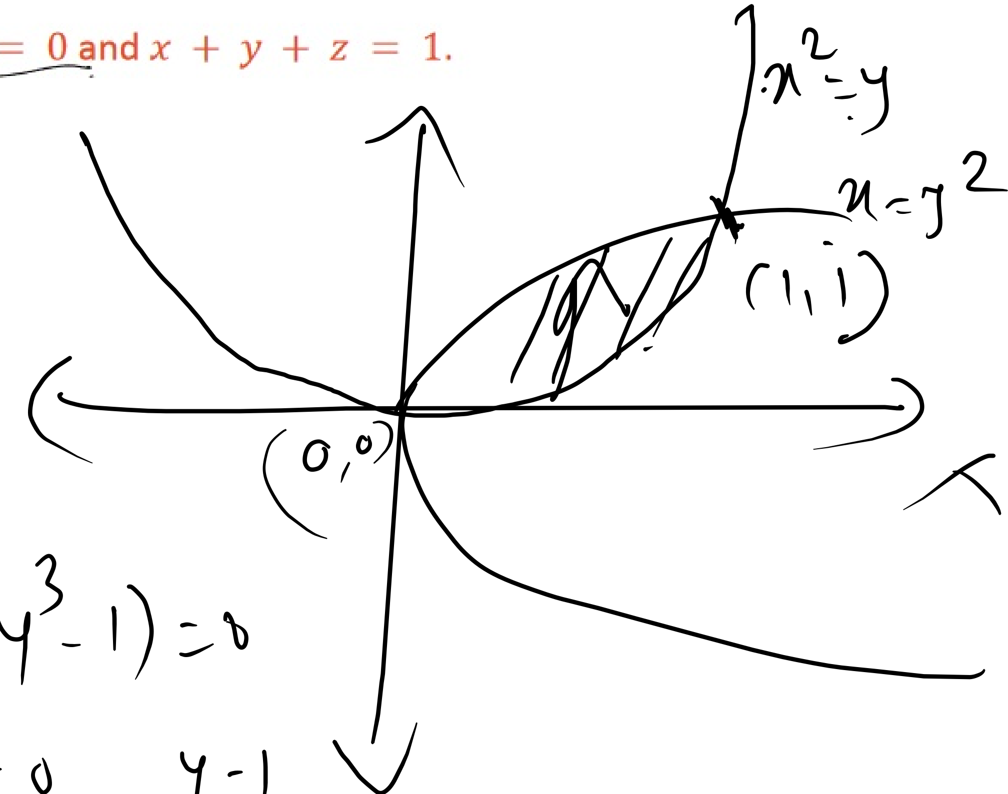
$$= \int_0^a \left[\frac{a^2}{2} - ax \cancel{-ax} \cancel{+ax} + \frac{x^2}{2} \right] dx$$

$$= \left[\frac{a^2}{2} x - \frac{ax^2}{2} + \frac{x^3}{6} \right]_0^a = \left[\cancel{\frac{a^3}{2}} - \cancel{\frac{a^3}{2}} + \frac{a^3}{6} \right]$$

2. Find the volume bounded by $y^2 = x, x^2 = y$ and the planes $z = 0$ and $x + y + z = 1$.

$x^2 = y$ $\qquad$ $x = y^2$

$x = \pm\sqrt{y}$ $\qquad$ $y = \pm\sqrt{x}$



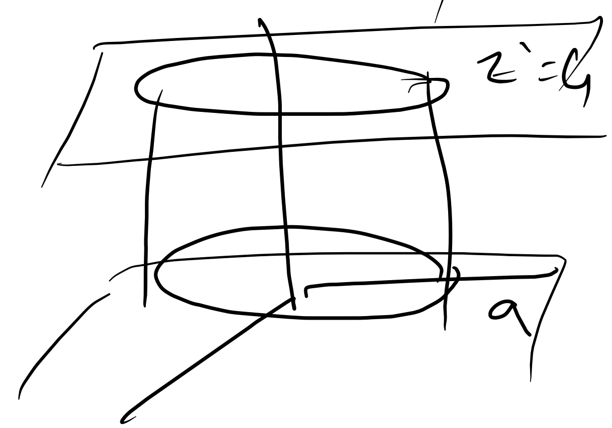
$y^4 = y$ $\qquad$ $y(y^3 - 1) = 0$

$y = 0 \quad y = 1$

$$V = \int_{x=0}^{1}\int_{y=x^2}^{\sqrt{x}}\int_{z=0}^{1-x-y} 1 \, dz\, dy\, dx$$

$$= \frac{1}{30}$$

★ $x^2+y^2=a^2$ $\qquad z=0 \qquad z=4$



$x = r\cos\theta$

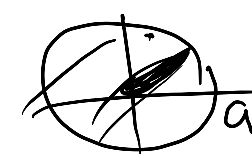
$y = r\sin\theta$

$z = z$

$dz\,dy\,dx = r\,dz\,dr\,d\theta$

$$V = \iiint dz\,dy\,dx = \int_{\theta=0}^{2\pi}\int_0^{a}\int_0^{4} r\,dz\,dr\,d\theta = \int^{2\pi}\int^{a} r(z)_0^4\,dr\,d\theta$$

or



$$= 4\int^{\pi/2}\int^{a}\int^{4} r\,dz\,dr\,d\theta$$

$=$

$$= \int^{2\pi}\int^{a} 4r\,dr\,d\theta$$

$$= 4\int^{2\pi}\left[\frac{r^2}{2}\right]_0^a d\theta$$

$$= \frac{\cancel{4}^{2}a^2}{\cancel{2}_{1}}\left[\theta\right]_0^{2\pi}$$

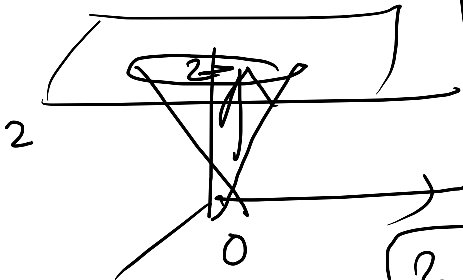
$$= 2a^2(2\pi)$$

$\pi a^2 4$

$x^2+y^2=z^2$

$z=0$ & $z=2$

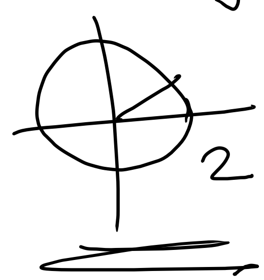
$x^2+y^2=2^2$ for $z=2$



$z: \sqrt{x^2+y^2} \to 2$

$V=\iiint dz\,dy\,dx$

$=\int_{\theta=0}^{2\pi}\int_{r=0}^{2}\int_{z=r}^{2} r\,dz\,dr\,d\theta$



$=\frac{1}{3}\pi\, 4\times 2$

$x=r\cos\theta$

$y=r\sin\theta$

$z=z$

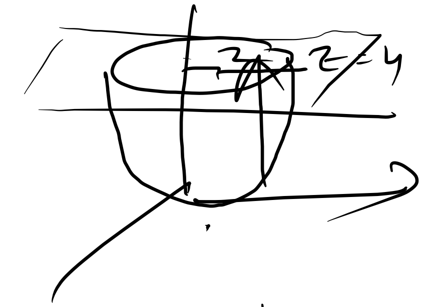
$x^2+y^2=z^2$

$r^2=z^2$

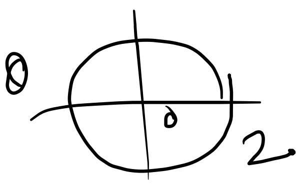
$x^2+y^2=z$

$z=0$ & $z=4$

$x^2+y^2=4$



$V=\int_0^{2\pi}\int_0^{2}\int_{r^2}^{4} r\,dz\,dr\,d\theta$



$z: x^2+y^2 \longrightarrow 4$

$: r^2 \to 4$

$=\int^{2\pi}\int^{2} r\,(z)_{r^2}^{4}\,dr\,d\theta$

$=\int^{2\pi}\int^{2} r\,(4-r^2)\,dr\,d\theta$

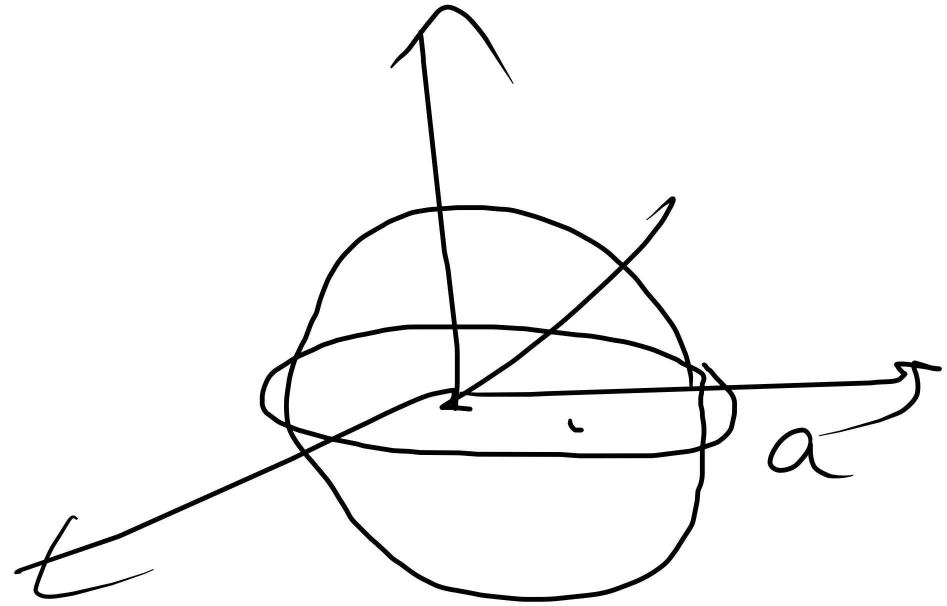
$=\int^{2\pi}\int^{2}(4r-r^3)\,dr\,d\theta$

$=\int^{2\pi}\left(\frac{4r^2}{2}-\frac{r^4}{4}\right)_0^2 d\theta$

$=(8-4)(\theta)_0^{2\pi}$

$=4(2\pi)=8\pi$

$$x^2+y^2+z^2=a^2$$

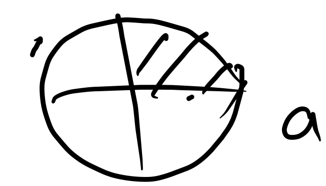


$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$z = r\cos\theta$$

$$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$$



$$\int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= 8\int_0^{\pi/2}\int_0^{\pi/2}\int_0^{a} r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$\frac{4}{3}\pi a^3$$

3. A cylindrical hole of radius b is bored through a sphere of radius a. Find the volume of the remaining solid.

$$\left.\begin{array}{l} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{array}\right\}$$

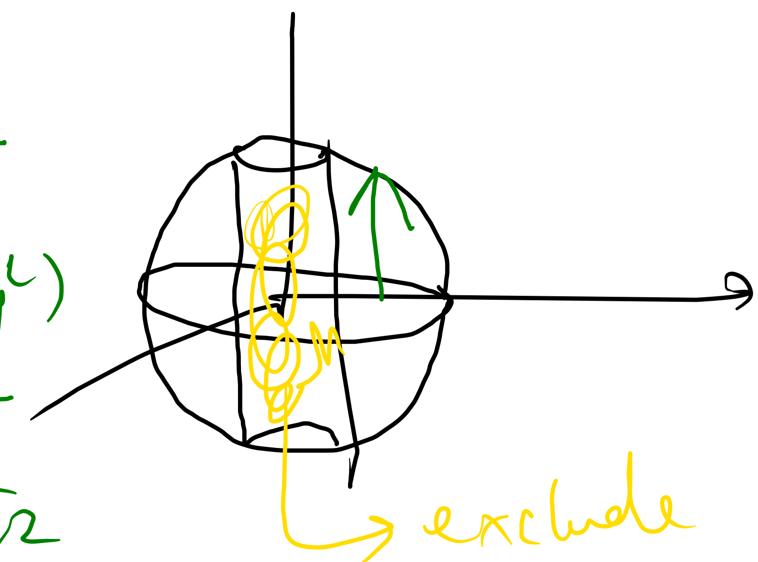
$dz\,dy\,dx = r\,dz\,dr\,d\theta$

$x^2 + y^2 + z^2 = a^2$

$z^2 = a^2 - (x^2 + y^2)$

$z^2 = a^2 - r^2$

$z = \pm\sqrt{a^2 - r^2}$



$$V = 8 \int_{\theta=0}^{\pi/2} \int_{b}^{a} \int_{z=0}^{\sqrt{a^2-r^2}} r\,dz\,dr\,d\theta$$

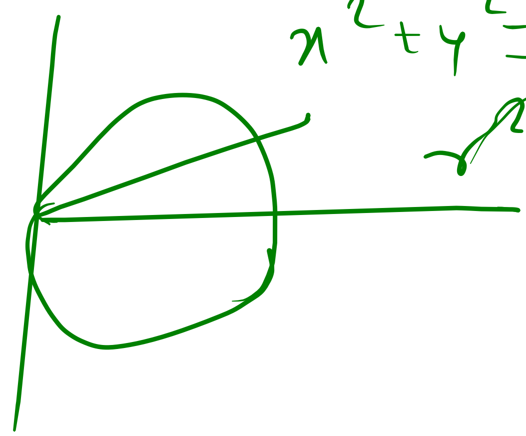
$$= 8 \int^{\pi/2} \int_b^a r\,(z)_0^{\sqrt{a^2-r^2}}\,dr\,d\theta$$

$$= 8 \int^{\pi/2} \int_b^a r\sqrt{a^2-r^2}\,dr\,d\theta$$

$a^2 - r^2 = t$

$$= \frac{4}{3}\pi (a^2 - b^2)^{3/2}$$

4. Show that the volume of the wedge intercepted between the cylinder $x^2 + y^2 = 2ax$ and planes $z = mx, z = nx$ is $\pi(n-m)\, a^3$.



$x^2 + y^2 = 2ax$

$r^{\cancel{2}} = 2a\cancel{r}\cos\phi$

5. Find the volume bounded by the cone $z^2 = x^2 + y^2$ and the paraboloid $z = x^2 + y^2$

$x = r\cos\theta$

$y = r\sin\theta$

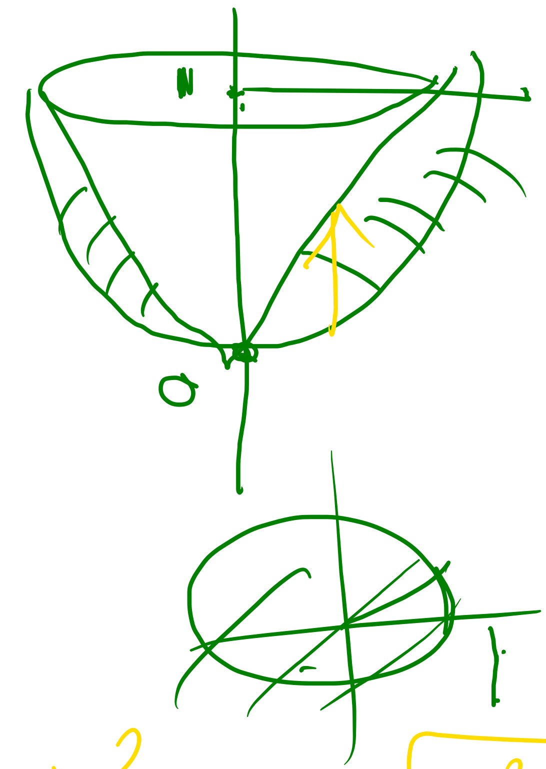
$z = z$

$dv = dy\,dx\,dz = r\,dz\,dr\,d\theta$

$z^2 = z$

$z(z-1) = 0$

$z = 0 \;,\; z = 1$



$$V = \int_0^{2\pi}\int_0^{1}\int_{r^2} r\,dz\,dr\,d\theta$$

$= \pi/6$

$z : x^2 + y^2 \longrightarrow \sqrt{x^2+y^2}$

$r^2 \longrightarrow r$

$z^2 = a^2 - r^2$

$x^2 + y^2 + z^2 = a^2$

$x^2 + y^2 = z^2$

$2z^2 = a^2$

$z^2 = a^2/2$

$x^2 + y^2 = (a/\sqrt{2})^2$

6. Find the volume cut off from the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$

$2z^2 = a^2$

$z^2 = \frac{a^2}{2}$

$x^2 + y^2 = \frac{a^2}{2}$

z: cone → sphere

$x = r\cos\theta$
$y = r\sin\theta$
$z = z$

$$V = \int_0^{2\pi}\int_0^{a/\sqrt{2}}\int_r^{\sqrt{a^2-r^2}} r\,dz\,dr\,d\theta$$

$$V = \int_0^{2\pi}\int_0^{a/\sqrt{2}} r\,(z)_r^{\sqrt{a^2-r^2}}\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{a/\sqrt{2}} \left(r\sqrt{a^2-r^2} - r^2\right) dr\,d\theta$$

$$= \int_0^{2\pi}\left[\int_0^{a/\sqrt{2}} r\sqrt{a^2-r^2}\,dr - \left(\frac{r^3}{3}\right)_0^{a/\sqrt{2}}\right] d\theta \quad \text{— (1)}$$

$$\int_0^{a/\sqrt{2}} r\sqrt{a^2-r^2}\,dr$$

$a^2 - r^2 = t$
$-2r\,dr = dt \Rightarrow r\,dr = -\frac{dt}{2}$
$r: 0 \to a/\sqrt{2}$
$t: a^2 \to a^2/2$

$$= \int_{a^2}^{a^2/2} t^{1/2}\left(\frac{-dt}{2}\right)$$

$$= \frac{1}{2}\int_{a^2/2}^{a^2} t^{1/2}\,dt = \frac{1}{2}\left(\frac{t^{3/2}}{3/2}\right)_{a^2/2}^{a^2} = \left[\frac{a^3}{3} - \frac{a^3}{3(2^{3/2})}\right]$$

from (1)

$$V = \int_0^{2\pi}\left[\frac{a^3}{3} - \frac{a^3}{3(2^{3/2})} - \frac{a^3}{2^{3/2}\,3}\right] d\theta$$

$2^{3/2}$

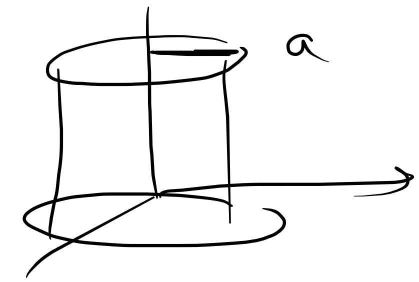
$(\sqrt{2})^3$

$$= \left(\frac{a^3}{3} - \frac{2a^3}{3 \cdot 2\sqrt{2}}\right)(\theta)_0^{2\pi}$$

$$= \frac{a^3}{3}\left(1 - \frac{1}{\sqrt{2}}\right)(2\pi)$$

$x^2+y^2=a^2$ $z=0$ $z=2$

$$V=\int_{\theta=0}^{2\pi}\int_{r=r}^{a}\int_{z=0}^{2} r\,dz\,dr\,d\theta$$



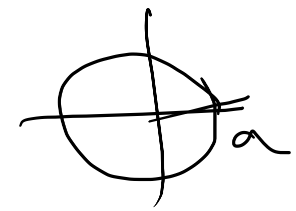
$x=r\cos\theta$

$y=r\sin\theta$

$z=z$

$dz\,dy\,dx = r\,dz\,dr\,d\theta$

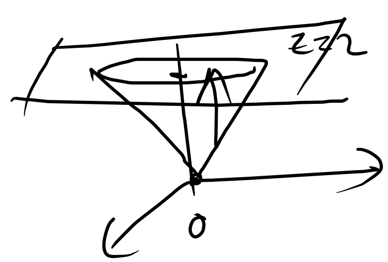
$+o^2(2)$



$x^2+y^2=z^2 \rightarrow$ cone

$z=0$ to $z=2$

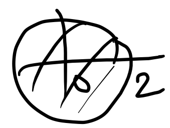
$$V=\int_0^{2\pi}\int_0^{2}\int_r^{2} r\,dz\,dr\,\theta$$



z: cone $\rightarrow$ plane

: $\sqrt{x^2+y^2}\rightarrow 2$

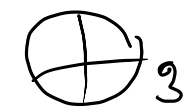
: $r\rightarrow 2$



volume

$x^2+y^2=z \rightarrow$ paraboloid

~~$z=0$~~ ~~to~~ $z=9$



z : parab. $\rightarrow$ plane

: $x^2+y^2\rightarrow 9$

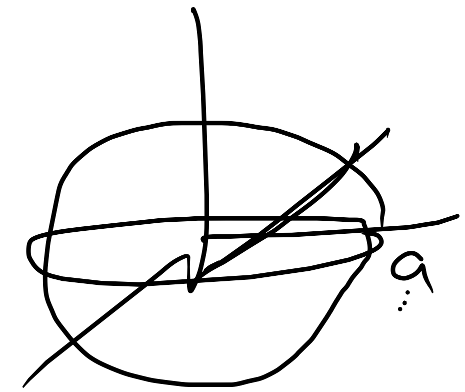
: $r^2\rightarrow 9$

$$V=\int_0^{2\pi}\int_0^{3}\int_{r^2}^{9} r\,dz\,dr\,d\theta$$

$\left(\frac{81}{2\times 2}-\frac{81}{4}\right)(2\pi)$ $\frac{81}{4}\times 2\pi$ $\frac{9r^2}{2}-\frac{r^4}{4}$ $9-r^2)$ $9r-r^3$

$x^2+y^2+z^2=a^2$

volume



Spherical

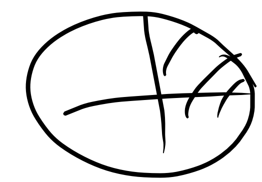
$x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

$\frac{4}{3}\pi a^3$

$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$



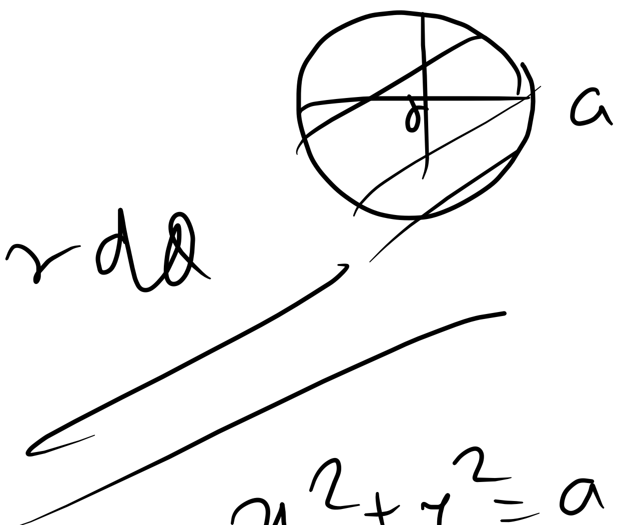
$$V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= 8\int_0^{\pi/2}\int_0^{\pi/2}\int_0^{a} r^2\sin\theta\,dr\,d\theta\,d\phi$$

7. Find the volume in the first octant bounded by the cylinder $x^2 + y^2 = 2$ and the planes $z = x + y$, $y = x, z = 0$ and $x = 0$.

8. Find the volume bounded by the paraboloid $x^2 + y^2 = az$ and the cylinder $x^2 + y^2 = a^2$.

$$\int_{0}^{2\pi}\int_{0}^{a}\int_{z=0}^{r^2/a} r\,dz\,dr\,d\theta$$



$x^2+y^2=az$

$r^2=az$

10. Find the volume cut off from the paraboloid $x^2 + \frac{1}{4}y^2 + z = 1$ by the plane $z = 0$.

11. Find the volume of the triangular prism formed by the planes $ay = bx, y = 0, x = a$ from $z = 0$ to $z = c + xy$

12. Find the volume of the solid that lies under the plane $3x + 2y + z = 12$ and above the rectangle $R\{(x, y)|0 \leq x \leq 1, -2 \leq y < 3\}$

$$\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$$

$$dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$x^2+y^2+z^2=a^2$$

$$V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} r^2\sin\theta\,dr\,d\theta\,d\phi$$